

MICHAEL MIRIERI

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PROFESSIONAL SUMMARY

Versatile Software Engineer and IT Professional with comprehensive expertise in full-stack web development, AI/ML engineering, relational database architecture, and cybersecurity infrastructure. Proven track record of delivering end-to-end applications—from neuro-symbolic AI health tech and custom database engines to secure e-commerce platforms and scalable IT solutions. Adept at cross-functional leadership, UI/UX design, and aligning technical execution with business strategy.

CORE SKILLS

- **Programming Languages:** Python, PHP, JavaScript, SQL, MySQL, PostgreSQL, HTML5/CSS3.
- **Frameworks & Libraries:** Next.js, React.js, Laravel, Django, Flask, Flutter, Tailwind.
- **AI, ML & Data:** RAG Architectures, LLMs (BioMistral v2), Agentic AI, NLP, OCR Integration, Data Analytics.
- **Security & Infrastructure:** CCNA, Ethical Hacking, Cryptography, Metasploit, Secure Coding, Linux.
- **Design & Architecture:** UI/UX Design, REST API Integration, Database Optimization, Enterprise Architecture.

PROFESSIONAL EXPERIENCE

Co-Founder & Technical Lead

FarmIQ | Nairobi, Kenya | Aug 2026 - Present

- Drafted functional specifications, system architecture requirements, and founder agreements for an agritech application.
- Designed operational workflows for farm logistics management utilizing QR code scanning technology.

ICT (Industrial Attachment)

Kenya Tea Development Agency (KTDA) | Jan 14, 2026 - Mar 11, 2026

- Supported cross-functional IT operations by assisting with cybersecurity monitoring, software development tasks, and data analytics to optimize organizational systems.

Freelance UI/UX Designer

Eldorado | 2025

- Designed user interfaces and mapped user journeys for a gaming website, focusing on intuitive navigation and highly engaging player experiences.

Freelance Software/Web Developer

Sarcom Center (SarcomLTD) | Aug 2024 - Sep 2024

- Designed and deployed a responsive business web application using HTML, CSS, and PHP, improving load performance by 25%.
- Gathered client requirements, tested features, and ensured stable deployment for commercial branding.

IT Support Assistant (Volunteer)

Kenya National Library Service (KNLS) | Jan 2024 - Mar 2024

- Spearheaded the implementation of a new bookkeeping system, leading a team to transition from manual to digital records.

KEY PROJECTS

SafeMedsAI — Neuro-Symbolic AI HealthTech App

- Developed an AI-powered medication safety platform combining OCR (to digitize medical labels) with a medical database to detect drug interactions.
- Implemented Retrieval-Augmented Generation (RAG) using BioMistral v2 to generate context-aware safety explanations (achieved 70% 'A' grade).

Mirieri DB (MDB) — Custom Relational Database Engine

- Engineered a lightweight relational database system in Python from scratch, featuring a custom SQL query parser and persistent storage logic.
- Optimized data retrieval efficiency to an $O(1)$ lookup time by implementing hash-map indexing for primary keys.

AI Chatbots & Virtual Assistants

- **Next.js Agentic AI Chatbot:** Architected a modern AI-driven conversational agent integrating multi-agent LLM workflows.
- **Python NLP Virtual Assistant:** Engineered a secure conversational agent incorporating NLP and strict input validation to mitigate injection vulnerabilities.

E-Commerce & Live Auction Platforms (BidUnity)

- Architected fully functional Laravel-based platforms featuring secure user authentication, product listings, and real-time live bidding logic.
- Optimized backend transaction workflows for highly concurrent bidding environments.

Pesapal API & Payment Gateway Integration

- Successfully integrated third-party payment gateway APIs (Pesapal JDEV26 Challenge), handling complex financial transaction workflows and data synchronization.

Property Management Architectures (Kejalink & Rental-Pro)

- Designed complex multi-entity relational database schemas linking user profiles, lease agreements, and financial transaction histories.

Cryptographic Password Manager

- Engineered a secure Password Manager applying rigorous cryptographic principles for safe credential storage and access control.

Beyond Phrases — Android Mobile Application

- Co-developed an Android mobile application designed for contextual phrase-based language learning.

EDUCATION

Strathmore University

Nairobi, Kenya | 2022 - 2026

Bachelor of Informatics and Computer Science — **Second Class Honours (Upper Division)**

Nairobi National High School

Nairobi, Kenya | 2018 - 2021

Kenya Certificate of Secondary Education (KCSE) — Grade: B+

CERTIFICATIONS

- IBM Data Science Badge
- Google AI Essentials
- Google Professional Data Analytics Certificate
- Google Professional Cybersecurity Certificate
- Cisco Certified Network Associate (CCNA) — Cisco
- Metasploit for Beginners: Ethical Penetration Testing